

Assignment # 2
Software Design & Architecture
BSE 4A (Spring 2025)

Deadline		
Group	As per Groups assigned for the Lab Project	
Submission Mode	LMS	
Total Marks	30	
CLO	CLO2, CLO3, CLO4	
CEP	WP1: Depth of Knowledge required	Apply knowledge from the domains of design patterns, quality tactics and UML. Correct application of patterns, quality tactics & UML will demonstrate your comprehension and ability to apply this knowledge in varying situations.
	WP3: Depth of Analysis Required	Have no obvious solution and require abstract thinking and originality in analysis to formulate suitable models. Alternative design solutions can exist to solve a particular design problem. Quality of design in terms of application of suitable architectural/design patterns will demonstrate your analysis.

Objective

You are required to create a software design specification document for the project you assigned to you (details mentioned below) for the courses of DBMS & SDA. You will follow the specifications provided to you for the database design including design constraints, database constraints and the type of database to be used. For this course, you will provide all relevant design artifacts that are required to build a working application that meets the given requirements and specified by you in the requirements specification.

Your Project has the following deliverables:

1. Requirement Specification
2. System Design Specification
3. A working application complying with the requirements and design specifications.

More specifically, the following tasks have to be performed.

A. Requirements Specification [5]

1. Develop requirements using User Stories categories as Features & Epics in Jira. Your specification report must conform to the relevant specification format. You must mention desired Non-functional requirements for each user story as well.

2. Elaborate the general requirements mentioned in the given project descriptions and make any assumptions as well as added features if required.
3. Share the link of Jira project by 15 May, 2025 by adding the following emails as Administrators
 - a. Awaismajeed.buic@bahria.edu.pk
 - b. asohail.buic@bahria.edu.pk

B. Design Specification

1. System Architecture and application of design patterns [5]

CLO3]

Select an appropriate architectural style/pattern to design your proposed system. Depict the selected system architecture as a block Diagram. Justify the Architectural Pattern/Style that has been used to design the system. Moreover, identify what kind of design patterns you will choose to meet the design constraints mentioned for your assigned project (descriptions provided in the project specifications).

2. Detailed Design [20]

[CLO2]

Produce detailed system design specification in the form of the following diagrams:

- a. Use Case Diagram showing important system features and actors interacting with the system. [2]
- b. UI Design of your application [2]
- c. Component Diagram presenting all important system components [2]
- d. Deployment Diagram presenting all important system components and their deployment in various computing nodes [2]
- e. Presentation of static view using Class Diagram [3]
- f. Data Model using appropriate modeling notation [2]
- g. Presentation of Dynamic View using
 - a. Sequence Diagram for all important use cases [3]
 - b. Activity Diagram for all important processes [2]
 - c. State transition diagram of the most important aspect (object, system, or subsystem) [2]

Design Constraints to be Considered:

Your project shall meet the following design constraints.

1. Application shall only allow access to authenticated users only.
2. Authenticated users shall only be able to perform authorized operations on the system.
3. Your system design shall support easy modification to the User interface during design and development phase by applying the principles of Separation of Concerns & Modularization. For this you will need to apply appropriate architectural patterns. Justify your choice.
4. All data needs to be stored in a central repository which can only be accessed through an appropriate application (Web, desktop, mobile etc.) to access and modify data. If modification to data or the way data is handled (business logic) needs to be modified, it shall have no effect on the User Interface (please refer to Constraint 3 discussed earlier).
5. The system shall be easily modifiable if some variation in the existing features has to be implemented without a major redesign.

Evaluation Criteria:

Evaluation criteria is as follows:

A(85 & above)	B(75-85)	C(60-74)	D(50-59)	F(less than 50)
- Comprehensive & Correct Requirements Specification - Correct Design Specification meeting all required & desired functional & Non-functional requirements. - OOD concepts are correctly implemented. - Correct architectural patterns/styles are applied and justification is given.	- Requirements specification meets essential requirements but has fewer mistakes and missing functional & non-functional requirements. - Design specification has some mistakes in UML models, system architecture but fundamental concepts have been correctly implemented. - No justification for architectural/design decision is given. - OOD concepts have been applied but correctness is not consistent.	- Requirements specification is incomplete, NFRs are not measurable and vague. - Design specification has mistakes with various views of UML are either missing or are not mapped with each other. - Wrong architectural & design pattern applied. - Fundamental UML & OOD concepts are applied incorrectly most of the times.	- Requirement specification document is incomplete and demonstrate poor understanding. - Distinction in Functional & non-functional requirements is made but lack consistency and correctness. - UML design is incomplete. - incorrect with few major mistakes in fundamental concept implementation. - OOD concepts have not been applied correctly.	- Incomplete and wrong specification both at functional and non-functional level. - UML diagrams are poorly drawn with no demonstration of usage of the correct diagrammatic notations and concepts. - OOD concepts are incorrectly applied demonstrating poor understanding.
- Engaging, Well focused, presentation	- Presentation covers all aspects but lacks	- Presentation is missing the required elements.	- Missing information - Poor time management	- Not presented - Missing information

	focus and able to finish hardly in time. - Lack bit of focus and communication skills are somewhat weak.	- Communication skills are average. - Presentation is cluttered with too much of information, diagrams are not readable	- Not engaging - Presentation/slide show has problems of content, readability	- Poor time management
- Developed application meets the specifications. - All design diagrams can be mapped correctly to the code. - Design patterns are applied correctly. - Project progress is depicted on Jira regularly.	- Developed application meets the specifications to considerable extent. - UML diagrams can be mapped to implementation artifacts most of the time with moderate level of violations. - Some of the Design patterns have been implemented to meet the constraints - Project progress is depicted on Jira.	- Developed application meets the specifications to considerable extent. - Some mapping of the UML diagrams to the code can be found. - Very few of the design patterns have been implemented - Project progress is depicted on Jira	- Developed application hardly meets the specifications. - Very little mapping of the UML diagrams to the code can be found. - No design patterns is implemented - The continuous project is depicted on Jira. - Project progress is depicted on Jira irregularly	- No working application. - No link of UML design with the implementation. - No pattern is applied. - Jira is not updated /all activities done at the end.
- Answers most of the questions correctly all the time demonstrating the knowledge about:	- Answers most of the questions correctly most of the time demonstrating the knowledge about:	- Answers most of the questions correctly most of the time demonstrating the knowledge about:	- Answers some of the questions correctly a shallow knowledge about: ○ Design patterns	- Unable to answer the fundamental questions demonstrating very weak or knowledge about:

○ Design patterns ○ Architectural patterns ○ Quality tactics ○ UML & OOD - Can strongly & correctly justify about the design choices made.	○ Design patterns ○ Architectural patterns ○ Quality tactics ○ UML & OOD - Can justify about the design choices made to some extent with some confidence.	○ Design patterns ○ Architectural patterns ○ Quality tactics ○ UML & OOD - Can justify about the design choices made to some extent with little confidence.	○ Architectural patterns ○ Quality tactics ○ UML & OOD - Can not justify about the design choices made to some extent. - Developed application hardly meets the specifications	○ Design patterns ○ Architectural patterns ○ Quality tactics ○ UML & OOD - Can not justify about the design choices made to some extent.
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